

Vedant Desai

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EDUCATION

University of Michigan

B.S. in Computer Science and Economics

Expected May 2028

Ann Arbor, MI

- **GPA:** 3.66 / 4.0

- **Coursework:** Data Structures & Algorithms, Intro to Statistics and Data Analysis, Discrete Mathematics, Calculus III

EXPERIENCE

CaseStudyPrep.AI

Dec 2025 – May 2026

Software Engineer Co-op (Voice AI)

Remote

- Moved audio processing off the UI thread into a Web Worker with an async stream handoff, reducing main-thread blocking time to under 5ms and keeping the real-time audio visualizer at a smooth 60 FPS during active inference.
- Eliminated a 27% audio upload failure rate with fault-tolerant RxJS logic that detects expired S3 presigned URLs, regenerates them mid-flight, and negotiates MIME types for WAV files Angular silently rejected.
- Eliminated the silent-audio problem in a voice-AI product — most frames sent to Whisper were dead air — by running Silero VAD client-side via ONNX Runtime, filtering silence before upload and cutting cloud inference costs by 40%.

Granular Synthesizer Plugin | C++, JUCE, VST3, AU

github.com/Verdent06/granular-synth

- Enforced the audio thread's zero-allocation constraint by pre-allocating a `MemoryPool<Grain, 64>` slab per voice at plugin startup — a fixed free-list that acquires and releases grain slots without touching the heap — so `processBlock()` never calls `new` or `delete` after `prepareToPlay()` completes.
- Wired UI-to-audio parameter delivery through a hand-rolled SPSC FIFO (64-slot fixed array, `std::atomic<int>` head/tail with acquire/release ordering) so slider changes reach the audio thread without a mutex — and routed WAV file handoff through a separate atomic shared_ptr swap, letting the UI thread publish a new sample without ever blocking `processBlock()`.
- Configured CMake to compile VST3 and AU plugin bundles from a single JUCE codebase — targeting a macOS universal binary (arm64 + x86_64) so the plugin runs natively on Apple Silicon and Intel — and audited every `processBlock()` code path against a real-time safety checklist confirming zero heap allocations and zero lock acquisitions before cutting release builds.

Michigan Data Consulting (MDC)

Jan 2026 – May 2026

Data Engineer — Michigan Campaign Finance Network

Ann Arbor, MI

- Replaced manual Michigan campaign-finance research — portal searches capped by the Bureau of Elections, irregular Excel exports, hand normalization at 2 hours per committee — with a Requests + Pandas ETL that ingests filings directly, eliminating 800 hours of manual pulls across 400 tracked PACs.
- Delivered a production Flask REST API on AWS EC2 as the sole engineer on a 5-month MCFN contract, wiring ingested data and PAC rankings into the nonprofit's public-facing research workflow.
- Scoped technical delivery directly with MCFN stakeholders as the only engineer on contract — from ingestion pipelines through REST endpoints on EC2 — with no backend team to share infrastructure, API design, or deployment ownership.

PROJECTS

SignalWeaver | FastAPI, React, TypeScript, PostgreSQL, Docker

github.com/Verdent06/SignalWeaver

- Containerized the stack with Docker Compose (API + Postgres/pgvector + nginx frontend) and a GitHub Actions CI pipeline (frontend build, pytest, API image build on main).
- Served composite financial-research scores through async REST endpoints (FastAPI) wrapping fundamentals, MPNet sentiment, and regression logic, instrumented end-to-end at 9.1s p50 / 15.2s p99 across 90 successful runs on 90 tickers.
- Built a React/TypeScript dashboard for composite scores and fundamental/sentiment/macro breakdowns; used it to analyze 90 distinct tickers with scores persisted to Postgres for history views.

TECHNICAL SKILLS

Languages Python, TypeScript, C++, SQL, HTML/CSS

Frameworks Angular, Flask, FastAPI, React, RxJS, JUCE

Cloud & Infrastructure AWS (EC2, S3), Docker, Git, GitHub Actions